PPAR Audit

Explain why reported portfolio performance changed.

PPAR Audit answers a difficult operational question: "Why did my reported portfolio performance change?" It compares two portfolio-accounting snapshots, quantifies supported causes across holdings, transactions, prices, FX, and related data, flags suspicious source-data relationships, and produces reviewer-ready Excel and HTML reports. When the available evidence is insufficient, the unexplained difference stays visible for human review.

- Everything runs locally, so portfolio data stays inside your environment.
- The Python implementation supports automated batch runs and local customization.
- Standard output includes XLSX, HTML, CSV, JSON, and compact evidence bundles.

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What PPAR Audit Answers

PPAR Audit is built around one operational question:

> **Why did my reported performance change?**

- **Performance Comparison:** identifies changed portfolio and security performance for each time period, quantitatively attributes defensible differences to supported source-data changes, and highlights anything that still needs human review.
- **Data Issues:** flags suspicious source-data relationships — including price ranges, dividend rates, accrued-interest rates, and missing dividends — that may indicate data-quality issues independently of the performance explanation.

Performance Auditing Report

Performance Differences

ROWS: 7

Portfolio	From Date	Thru Date	Performance Difference	Explained Difference	Unexplained Difference	Status	Comments
ALPHA	2026-01-31	2026-02-27	0.001495	0.001495		**Fully Explained**	
ALPHA	2026-05-01	2026-05-29	0.002528	0.002528		**Fully Explained**	
BALANCED	2026-04-01	2026-04-30	-0.002138	-0.002138		**Fully Explained**	
BALANCED	2026-05-09	2026-05-14	-0.007289	-0.007489	0.000200	**Partly Explained**	Possible cause: JPM transactions.amount increased by 200.00 on 2026-05-12. Add YAML configuration to count it as explained.
INCOME	2026-01-01	2026-01-30	-0.000050	-0.000050		**Fully Explained**	
INCOME	2026-02-28	2026-03-31	0.000005	0.000005		**Fully Explained**	
INCOME	2026-04-01	2026-04-30	0.000350	0.000000	0.000350	**Unexplained**	The Unexplained Difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications.

Performance Difference Causes

ROWS: 25

Portfolio	From Date	Thru Date	As Of Date	Dataset.Field	Security	Snapshot A Value	Snapshot B Value	B - A Difference	Performance Difference Explained	Explanation
ALPHA	2026-01-31	2026-02-27	2026-02-10	transactions.amount	91282Y5Y1		-8.000000	-8.000000		pa: Caused cash-balance ending holdings.market_value to decrease by 8.00.
ALPHA	2026-01-31	2026-02-27	2026-02-17	transactions.amount	CASH_USD		-2000.000000	-2000.000000	0.002001	lo: External flow decreased by 2,000.00; weighted external flow decreased by 785.71.
ALPHA	2026-01-31	2026-02-27	2026-02-27	holdings.market_value	CASH_USD	16000.000000	13992.000000	-2008.000000	-0.002008	CASH_USD ending holdings.market_value decreased by 2,008.00.
ALPHA	2026-01-31	2026-02-27	2026-02-27	holdings.quantity	CASH_USD	16000.000000	13992.000000	-2008.000000		CASH_USD ending holdings.quantity decreased by 2,008.00.
ALPHA	2026-05-01	2026-05-29	2026-04-30	holdings.market_value	CASH_USD	16000.000000	16537.010000	537.010000	-0.000538	CASH_USD beginning holdings.market_value increased by 537.01.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.market_value	AAPL	163969.350000	165535.380000	1566.030000	0.001566	AAPL ending holdings.market_value increased by 1,566.03.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.market_value	CASH_USD	16000.000000	17500.000000	1500.000000	0.001500	CASH_USD ending holdings.market_value increased by 1,500.00.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.price	AAPL	161.000000	162.610000	1.610000		AAPL ending holdings.price increased by 1.61.
ALPHA	2026-05-01	2026-05-29	2026-05-29	holdings.quantity	CASH_USD	16000.000000	17500.000000	1500.000000		CASH_USD ending holdings.quantity increased by 1,500.00.
BALANCED	2026-04-01	2026-04-30	2026-04-06	transactions.amount	JPM	469.880000	586.950000	117.070000		dv: Caused cash-balance ending holdings.market_value to increase by 117.07.

BALANCED	2026-04-01	2026-04-30	2026-04-18	transactions.amount	JPM		240.000000	240.000000		rc: Caused cash-balance ending holdings.market_value to increase by 240.00.
BALANCED	2026-04-01	2026-04-30	2026-04-30	holdings.market_value	CASH_USD	64000.000000	64357.070000	357.070000	0.000357	CASH_USD ending holdings.market_value increased by 357.07.
BALANCED	2026-04-01	2026-04-30	2026-04-30	holdings.quantity	CASH_USD	64000.000000	64357.070000	357.070000		CASH_USD ending holdings.quantity increased by 357.07.
BALANCED	2026-05-09	2026-05-14	2026-05-08	holdings.market_value	AAPL	161823.820900	163194.090000	1370.269100	-0.001360	AAPL beginning holdings.market_value increased by 1,370.27.
BALANCED	2026-05-09	2026-05-14	2026-05-08	holdings.market_value	CVNA	1708.760000	8204.920000	6496.160000	-0.006447	CVNA beginning holdings.market_value increased by 6,496.16.
BALANCED	2026-05-09	2026-05-14	2026-05-12	transactions.amount	JPM		200.000000	200.000000		rc: Caused cash-balance ending holdings.market_value to decrease by 200.00. Add YAML configuration to count it as explained.
BALANCED	2026-05-09	2026-05-14	2026-05-14	holdings.market_value	MSFT	109052.630000	109370.040000	317.410000	0.000317	MSFT ending holdings.market_value increased by 317.41.
BALANCED	2026-05-09	2026-05-14	2026-05-14	holdings.quantity	MSFT	956.602000	959.386200	2.784200		MSFT ending holdings.quantity increased by 2.78.
INCOME	2026-01-01	2026-01-30	2026-01-20	transactions.amount	CASH_USD	-50.000000	-100.000000	-50.000000		dp: Caused cash-balance ending holdings.market_value to decrease by 50.00.
INCOME	2026-01-01	2026-01-30	2026-01-30	holdings.market_value	CASH_USD	128000.000000	127950.000000	-50.000000	-0.000050	CASH_USD ending holdings.market_value decreased by 50.00.
INCOME	2026-01-01	2026-01-30	2026-01-30	holdings.quantity	CASH_USD	128000.000000	127950.000000	-50.000000		CASH_USD ending holdings.quantity decreased by 50.00.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.accrued	91282Y5Y1	4.560000	4.580000	0.020000	-0.000000	91282Y5Y1 beginning holdings.accrued increased by 0.02.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.market_value	91282Y5Y1	48000.000000	48197.600000	197.600000	-0.000198	91282Y5Y1 beginning holdings.market_value increased by 197.60.
INCOME	2026-02-28	2026-03-31	2026-02-27	holdings.market_value	CASH_USD	128000.000000	127797.150000	-202.850000	0.000203	CASH_USD beginning holdings.market_value decreased by 202.85.
INCOME	2026-04-01	2026-04-30								No identifiable cause was found. Review `supporting_files/source_detail.csv`. The difference may be due to missing source-data, source-file timing differences, or vendor methodology that does not match the YAML specifications.

Data Audit Issues

ROWS: 6

Snapshot	Portfolio	As Of Date	Dataset.Field	Security	Issue Type	Reference Value	Observed Value	Difference	Tolerance	Explanation
Snapshot A	ALPHA	2026-05-23	transactions.amount	AAPL	missing_dividend				same-date dividend present	Missing a dividend for AAPL on 2026-05-23 that is in portfolio INCOME.
Snapshot B	ALPHA	2026-02-10	transactions.amount	91282Y5Y1	pa_sa_rate	0.086588	0.131733	0.045145	greater of 0.000001 or 0.5%	pa: Same-day same-security rate differs across portfolios for 91282Y5Y1.
Snapshot B	ALPHA	2026-05-29	holdings.price	AAPL	holdings_price_range	161.000000	162.610000	1.610000	greater of 0.01 or 0.05%	Same-day same-security holdings.price differs across portfolios for AAPL.
Snapshot B	BALANCED	2026-04-06	transactions.amount	JPM	dividend_rate	1.399998	1.748792	0.348794	greater of 0.01 or 0.5%	dv: Same-day same-security rate differs across portfolios for JPM.
Snapshot B	INCOME	2026-01-30	holdings.accrued	91282Y5Y1	holdings_accrued_rate	0.009386	0.010209	0.000823	greater of 0.00001 or 0.5%	Same-day same-security holdings.accrued per unit differs across portfolios for 91282Y5Y1.
Snapshot B	INCOME	2026-03-05	transactions.price	AAPL	transactions_price_range	161.000000	161.150000	0.150000	greater of 0.01 or 0.05%	Same-day same-security transactions.price differs across portfolios for AAPL.

Setup

Install the PPAR package:

```
pip install ppar
```

Create a local PPAR Audit workspace. The workspace includes PPAR-normalized demonstration data modeled on Axys/APX source and report data, so you can run the complete workflow before replacing the CSV files with reviewed exports from your own environment.

```
ppar setup ./my_ppar_audit
```

Run Audit:

```
ppar audit ./my_ppar_audit
```

Follow the `Customizing With Your Own Data` section in `./my_ppar_audit/README.md` when you are ready to customize the workspace with your own data.

```
my_ppar_audit/
  README.md
  ppar.yaml
  run_audit.py
  snapshot_a/
    portperf.csv
    holdings.csv
    transactions.csv
    secmast.csv
    secperf.csv
    fx_rates.csv
```

```
    splits.csv
  snapshot_b/
    portperf.csv
    holdings.csv
    transactions.csv
    secmast.csv
    secperf.csv
    fx_rates.csv
    splits.csv
```

Inputs

The demonstration files are PPAR-normalized CSVs modeled on Axys/APX data. Production inputs may come from reviewed REP, IMEX, custom-report, or other controlled exports:

- portfolio performance;
- security performance;
- holdings;
- transactions;
- security master data;
- FX rates; and
- split factors.

Audit uses two source-data snapshots. Snapshot A is normally the older or original state and Snapshot B is normally the newer or restated state, but neither snapshot is presumed correct.

PPAR normalizes those files through a customizable `ppar.yaml` file, so each site can configure its local field names, transaction-code treatment, comparison tolerances, and report assumptions. The setup-created file documents these choices in place.

The configured file and accounting contracts fail closed when required source treatment is missing or ambiguous. Optional evidence does not silently expand the calculation or policy surface.

Outputs

PPAR Audit writes review packages:

```
output/
  portfolio/
    portfolio_audit.xlsx
    portfolio_audit.html
    source_detail.csv
    audit_support.zip
  security/
    security_audit.xlsx
    security_audit.html
    source_detail.csv
    audit_support.zip
```

To prevent unusably large artifacts, Audit stops with a nonzero exit code before writing a report when any primary review table would exceed 100,000 rows. The error identifies the oversized table and its largest contributors so the user can narrow the portfolio or date scope or correct upstream differences.

Current Validation Scope

PPAR Audit has substantial automated coverage, financial invariants, report reconciliation checks, output-integrity checks, deterministic demonstrations, and maintained scale gates.

It has not yet been validated against a real client's production-style Axys/APX exports and approved local accounting policy. The current program is seeking a small number of strong validation partners to test source authenticity, setup burden, financial interpretation, false positives, and reviewer usefulness.

PPAR Audit detects, compares, explains, and helps investigate supported portfolio-performance and source-data differences. It does not provide a financial-statement audit, GIPS verification, attestation, certification, or assurance opinion.

Additional Repository Capability

This repository also contains `ppar.analytics`, a maintained module for benchmark-relative performance attribution, contribution, and ex-post risk reporting. It is retained for future PPAR packaging but is not part of the current PPAR Audit validation program or default onboarding workflow.